GENERATIVE AI COURSE

Module 3: Generative AI Tools & Real-World Applications

Learning Objectives

- Understand different categories of Generative AI tools and their applications.
- Learn how to generate text, images, presentations, code, and other digital content using AI.
- Understand multimodal AI and how it works with text, images, audio, and video.
- Learn to use AI tools for education, software development, business, and content creation.
- Practice completing real-world tasks using multiple AI tools.
- Learn to verify AI-generated information and use AI responsibly.

1. Topics

Introduction to Generative AI Tools

- What are Generative AI tools?
- How are AI tools different from traditional software?
- Popular applications of Generative AI.
- Choosing the right AI tool for a specific task.

Examples: ChatGPT, Google Gemini, Microsoft Copilot, Claude, Canva AI, and GitHub Copilot.

AI Tools for Text Generation

- Generating articles, notes, emails, and reports.
- Summarizing long documents.

- Rewriting and proofreading content.
- Translating text into different languages.
- Generating quizzes and study material.

Example: Ask AI to convert a long technical chapter into simple revision notes for beginners.

AI Tools for Image Generation

- Introduction to text-to-image generation.
- Creating images using descriptive prompts.
- Understanding image style, subject, composition, lighting, and background.
- Generating posters, illustrations, advertisements, and social media graphics.
- Editing existing images using AI tools.

Example: Generate a promotional poster for a college coding competition.

AI Tools for Presentation Creation

- Creating presentation outlines using AI.
- Generating slide content.
- Converting topics into presentation structures.
- Designing presentations using tools such as Canva AI.
- Improving slide titles, layouts, and visual storytelling.

Example: Create a 10-slide presentation on Artificial Intelligence for a beginner audience.

AI Tools for Coding and Software Development

- Generating code from natural-language instructions.
- Explaining existing code.
- Debugging errors.
- Generating test cases.
- Improving code readability.
- Creating documentation and README files.

Example: Ask AI to create a student registration form using HTML, CSS, and JavaScript, then test and improve the code.

AI Tools for Data Analysis

- Understanding datasets.
- Generating SQL queries and Python code.
- Summarizing CSV and Excel data.
- Identifying trends and patterns.
- Creating charts and reports.
- Understanding the limitations of AI-generated analysis.

Example: Analyze student marks and identify the highest score, average marks, and students needing revision.

Introduction to Multimodal AI

- What is multimodal AI?
- Text-to-image and image-to-text tasks.
- Speech-to-text and text-to-speech.
- Working with audio and video.
- Using uploaded images and documents as AI inputs.
- Practical applications of multimodal AI.

AI for Education and Learning

- Generating personalized study plans.
- Creating topic-wise quizzes.
- Explaining difficult concepts using real-life examples.
- Practicing interview questions.
- Creating revision notes and flashcards.
- Learning with AI without becoming dependent on generated answers.

AI for Business and Productivity

- Writing professional emails and reports.
- Creating marketing content.
- Generating business ideas.
- Preparing customer support responses.
- Organizing schedules and task lists.
- Automating repetitive writing and documentation tasks.

AI for Social Media and Content Creation

- Generating Instagram captions and post ideas.
- Creating content calendars.
- Writing video scripts and reels captions.
- Generating visual concepts for advertisements.
- Adapting content for different audiences and platforms.

AI Output Verification and Responsible Use

- Why AI-generated responses can be incorrect.
- Fact-checking important information.
- Identifying fabricated references and unsupported claims.
- Protecting personal and confidential information.
- Copyright, attribution, and responsible content use.
- Reviewing generated code before running it.
- Using AI as an assistant rather than blindly trusting its output.

2. Practical Exercises

Practical 21: AI Content Creator

Ask ChatGPT:

> **"Act as a professional content writer. Create a 300-word article explaining the benefits of learning Python to college students. Use simple English, headings, and practical examples."**

Follow-up task: Convert the article into five short social media captions.

Learning: Text generation, role-based prompting, and content repurposing.

Practical 22: AI Image Generator

Use an AI image-generation tool.

Prompt:

> **"Create a modern promotional poster for a college-level coding competition. Include a computer screen, programming symbols, students collaborating, a professional layout, and a technology-themed background."**

Follow-up task: Generate a second version with a minimalist design.

Learning: Image generation and visual prompt refinement.

Practical 23: AI Presentation Creator

Ask AI:

> **"Create a 10-slide presentation outline on Generative AI for beginner students. Include slide titles, key points, real-life examples, and a concluding slide."**

Follow-up task: Use a presentation tool to create slides from the outline.

Learning: Presentation planning and AI-assisted design.

Practical 24: AI Coding Assistant

Ask ChatGPT:

> **"Act as a frontend developer. Create a responsive student registration form using HTML, CSS, and JavaScript. Include name, email, phone number, course, and a submit button. Validate required fields and explain the code for beginners."**

Follow-up task: Test the code, find a validation issue, and ask AI to fix it.

Learning: Code generation, debugging, and testing.

Practical 25: AI SQL Assistant

Ask AI:

> **"Create a MySQL database table named students with student_id, name, course, and marks. Insert five sample records, then write queries to display all students, find the highest marks, calculate average marks, and identify students scoring above 80."**

Follow-up task: Run each query in MySQL and verify the results.

Learning: AI-assisted SQL generation and validation.

Practical 26: AI Data Analyst

Provide a small student marks dataset to an AI tool capable of analyzing files.

Prompt:

> **"Analyze this student marks dataset. Calculate the average marks, identify the highest and lowest marks, count students scoring below 40, and summarize the results in a table. Explain any assumptions."**

Follow-up task: Verify the numerical calculations using Python or Excel.

Learning: Data analysis and result verification.

Practical 27: AI Study Assistant

Ask AI:

> **"Act as a Python instructor. Create a study plan for a beginner who wants to learn Python in 7 days. Include daily topics, explanations, coding exercises, and a short quiz for each day."**

Follow-up task: Ask AI to make the plan easier for a student who has only 30 minutes per day.

Learning: Personalized learning and iterative improvement.

Practical 28: AI Business Content Creator

Prompt:

> **"Act as a digital marketing assistant. Create a marketing campaign for a new online coding course. Include a target audience, three Instagram captions, five hashtags, one advertisement headline, and a call to action."**

Follow-up task: Rewrite the campaign in a more professional tone.

Learning: AI-assisted marketing and business communication.

Practical 29: Multimodal AI Challenge

Provide an image of a product or a suitable classroom photograph to an AI tool that supports image input.

Prompt:

> **"Describe the visible elements in this image. Suggest three suitable social media captions and explain which audience each caption would appeal to. Do not assume details that are not visible."**

Follow-up task: Compare the AI description with the actual image and identify any incorrect assumptions.

Learning: Image understanding, contextual prompting, and output verification.

Practical 30: AI Fact-Checking Challenge

Prompt:

> **"Explain five important facts about Generative AI and provide reliable sources that can be used to verify each claim."**

Student tasks:

1. **Identify five factual claims in the response.**
2. **Locate reliable sources for each claim.**

3. Check whether the sources actually support the claims.
4. Correct inaccurate or unsupported information.
5. Submit the original response and the verified version.

Learning: Fact-checking, research skills, and responsible AI usage.

3. Classroom Challenge

Challenge: Build a Complete AI-Assisted Project

Students must create a promotional campaign for a fictional technology workshop.

The submission must include:

1. An event description generated using AI.
2. A promotional poster created with an AI image tool.
3. Five social media captions.
4. A five-slide presentation outline.
5. A final review identifying and correcting any inaccurate or unsuitable AI-generated content.

Evaluation criteria: Prompt quality, creativity, relevance, output quality, verification, and originality.

4. Practical Workflow

Understand the task → Select an AI tool → Write the prompt → Generate content → Review the output → Verify important details → Refine the result → Submit the final work

5. Homework / Assignment

- **Use two different AI tools to explain the same technical topic and compare their responses.**
- **Generate one AI image using a detailed prompt.**
- **Create a five-slide presentation outline on a technical topic.**
- **Use AI to generate a small Python or JavaScript program, then test it.**
- **Analyze a sample dataset using AI and verify the results.**
- **Identify three possible errors or limitations in AI-generated content.**
- **Write a short report explaining which AI tool was most useful for each task and why.**

Submission Requirements

- **Student name and GRID number.**
- **Prompts used for each activity.**
- **AI-generated outputs or screenshots.**
- **Corrections and verification performed by the student.**
- **A short conclusion describing what was learned.**

Expected Outcome: Students will be able to select appropriate Generative AI tools, use them for practical tasks, refine generated content, and verify results before using them.